

Flowing before decimating leaves every norm-based criterion unchanged: what is needed is a statement about the state

Conjugating by the flow before blocking does not improve the criterion of the blocking-criterion note, and the reason is one line: Φ_t is a bijection of the configuration space, so

$$\|V \circ \Phi_t\|_\infty = \sup_U V(\Phi_t(U)) = \sup_{U'} V(U') = \|V\|_\infty,$$

even though $V(\Phi_t(U)) \leq V(U)$ holds pointwise by the monotonicity of the flow. The straddling-plaquette norm that enters β_{block} is therefore unchanged, the block gap is unchanged because conjugation is unitary, and the factor $3M^2/8$ by which blocking loses is unchanged. The same argument disposes of every criterion built from operator norms: **no reordering of conjugation and decimation can help, because the quantities such a criterion uses are conjugation-invariant or bijection-invariant.** The flow's gain is pointwise and therefore visible only in expectation, that is, only against a state whose weight sits where the flowed action is small. With that, the last inexpensive idea in this programme is closed, and the residual statement is sharp: the remaining input is control of the ground-state measure, which is the constructive problem in its Euclidean form. Constants explicit; nothing promoted.

1. The three invariances

Let Φ_t be the lattice flow map and W_t its unitary implementation, as in the flow-conjugation note Proposition 1.

(i) **Spectra are conjugation-invariant.** Every eigenvalue of H , every gap, and in particular the block gap Δ_M of a block Hamiltonian, is unchanged by $H \mapsto W_t H W_t^*$.

(ii) **Sup norms of multiplication operators are bijection-invariant.** For $f \in C(\mathcal{M})$ and a bijection Φ of $\mathcal{M}$, $\|f \circ \Phi\|_\infty = \|f\|_\infty$. Applied to the magnetic term,

$$\|V \circ \Phi_t\|_\infty = \|V\|_\infty = \frac{4N\hbar c}{ag^2} \cdot \#\{\text{plaquettes}\},$$

and to the straddling part, $\|\sum_{p \text{ str}} w_p \circ \Phi_t\|_\infty$ is bounded by the same $6M^2 \cdot 4N\hbar c/(ag^2)$ per block as before. The pointwise inequality $V \circ \Phi_t \leq V$, which the monotonicity of the flow supplies, is compatible with equality of the suprema because Φ_t maps the configuration space onto itself.

(iii) **Hence the criterion is unchanged.** The blocking criterion

$$\beta_{\text{block}} = \frac{\|\sum_{p \text{ str}} w_p\|_\infty}{\Delta_M} = \frac{24M^2 N}{g^2 \delta(g; M)}$$

has a numerator fixed by (ii) and a denominator fixed by (i), so conjugating before blocking leaves it exactly as computed in the blocking-criterion note, and blocking still loses by $3M^2/8$.

2. Why no reordering helps

Any criterion of the form (perturbation norm)/(unperturbed gap) is built from two quantities of which the first is invariant under bijections of the configuration space and the second under unitary conjugation. The flow supplies exactly a bijection and a unitary. So:

Proposition. Let $\mathcal{C}(H_0, \phi)$ be any criterion depending on H_0 only through its spectrum and on ϕ only through $\|\phi\|_\infty$. Then $\mathcal{C}$ takes the same value for (H_0, ϕ) and for $(W_t H_0 W_t^*, \phi \circ \Phi_t)$, for every flow time t .

This covers the Yarotsky criterion used in the T2 note, its blocked version, and the Schur-complement criteria of the Feshbach note, all of which reduce to comparisons of norms with gaps.

3. Where the flow's gain actually lives

The flow does reduce the action: $V(\Phi_t(U)) \leq V(U)$ for every U , with strict inequality away from critical points, and $\frac{d}{ds}S(B_s) = -\|D^*G\|_2^2$. That gain appears in expectations,

$$\langle \psi, (V \circ \Phi_t) \psi \rangle \leq \langle \psi, V\psi \rangle \quad \text{for every state } \psi,$$

and is large when the weight of ψ sits where the flow moves the configuration far, that is, on rough configurations. So the flow converts a rough state into a smooth one at the level of expectations while leaving every worst-case bound alone.

A criterion that could see the gain must therefore be **relative to the state**, of the form

$$\langle \psi, \phi \psi \rangle \leq \epsilon \langle \psi, (H - E_0) \psi \rangle + \eta \|\psi\|^2 \quad \text{on the low-energy subspace,}$$

which is the relative form already isolated in the Schur note Lemma 1' and in the relative hypothesis of the Feshbach note §3a. The quantity it needs is the distribution of the field strength in the low-energy states, that is, the ground-state measure $|\Omega(U)|^2 dU$, which is a probability measure on the configuration space and is exactly the object the Euclidean formulation supplies explicitly as $e^{-S}dU$ and the Hamiltonian formulation leaves implicit.

4. The residual statement

Collecting the closed routes of this programme:

method	status	reason
variational upper bounds	closed for $m > 0$	a spectral measure with all negative moments finite can have $m = 0$
expansion around the free theory	closed	bound reaches m only at the confinement scale
blocking by projection	closed	boundary grows like M^2 , block gap does not
conjugation by the flow	closed	spectrum-invariant; the gain is the truncation, which is cheap
flow before decimation	closed (this note)	norms are bijection-invariant
decimation with controlled couplings	open	the constructive problem

Every method that uses only norms and spectra is exhausted, and each was closed by a computation with explicit constants. What is left needs the distribution of the field in the low-energy states, and that is where the Euclidean construction has its tools and the Hamiltonian formulation has none of its own.

5. Consequence for STATE

The question raised at the end of the lattice-truncation note is answered in the negative, with the invariance that makes it so. The programme's norm-based line is complete. The next thing that would advance it is an estimate on the ground-state measure of the Kogut–Susskind Hamiltonian at intermediate coupling, for instance a bound of the form $|\Omega(U)|^2 \leq C e^{-\lambda S_w(U)}$ with $\lambda > 0$ uniform in the volume, which would import the Euclidean large-field machinery into the Hamiltonian setting. That is a well-posed question and it is the natural successor to everything above.